focusing on electrons, the uncertainty principle is completely general: it applies to everything.

In day-to-day life we routinely speak about things like a car passing a particular stop sign (position) while traveling at 90 miles per hour (velocity), blithely specifying these two physical features. In reality, quantum mechanics says that such a statement has no precise meaning since you can't ever simultaneously measure a definite position and a definite speed. The reason we get away with such incorrect descriptions of the physical world is that on everyday scales the amount of uncertainty involved is tiny and generally goes unnoticed. You see, Heisenberg's principle does not just declare uncertainty, it also specifies —with complete certainty—the minimum *amount* of uncertainty in any situation. If we apply his formula to your car's velocity just as it passes a stop sign whose position is known to within a centimeter, then the uncertainty in speed turns out to be just shy of a billionth of a billionth of a billionth of a billionth of a mile per hour. A state trooper would be fully complying with the laws of quantum physics if he asserted that your speed was between 89.99999999999999999999999999999999 and 90.00000000000000000000000000000001 miles per hour as you blew past the stop sign; so much for a possible uncertainty-principle defense. But if we were to replace your massive car with a delicate electron whose position we knew to within a billionth of a meter, then the uncertainty in its speed would be a whopping 100,000 miles per hour. Uncertainty is always present, but it becomes significant only on microscopic scales.

The explanation of uncertainty as arising through the unavoidable disturbance caused by the measurement process has provided physicists with a useful intuitive guide as well as a powerful explanatory framework in certain specific situations. However, it can also be misleading. It may give the impression that uncertainty arises only when we lumbering experimenters meddle with things. This is not true. Uncertainty is built into the wave structure of quantum mechanics and exists whether or not we carry out some clumsy measurement. As an example, take a look at a particularly simple probability wave for a particle, the analog of a gently rolling ocean wave, shown in Figure 4.6. Since the peaks are all uniformly moving to the right, you might guess that this wave describes a particle moving with the velocity of the wave